The Campus Network Lockdown: A Ransomware Case Study

What the crime is: Ransomware is a type of malicious software that encrypts a victim's files or completely locks them out of their own computer systems. The cybercriminals then demand a ransom payment—usually in an untraceable cryptocurrency—in exchange for the decryption key to restore access.

How it happens (Step-by-Step):

1. **Infiltration:** A university staff member receives a highly realistic phishing email appearing to be from the campus IT department, urging them to download a "mandatory security patch."
2. **Execution:** The staff member clicks the link and downloads the file. The malware installs silently in the background.
3. **Encryption:** The ransomware rapidly moves laterally through the university's network, hunting down and encrypting critical servers containing student records, research databases, and financial data.
4. **Extortion:** System administrators suddenly find themselves locked out. A digital ransom note appears on all affected screens, demanding $200,000 in Bitcoin within 48 hours, threatening to permanently delete or leak the data online if ignored.

Target Demographic: While anyone can be a victim, ransomware gangs specifically target institutions like universities, hospitals, and local governments. These organizations desperately need their data to function day-to-day and often have vulnerable, outdated cybersecurity infrastructure.

Consequences: The fallout is devastating. Institutions suffer massive operational paralysis, severe financial losses from recovery efforts (even if the ransom isn't paid), and permanent reputational damage. If personal data is stolen, students and staff are put at long-term risk of identity theft.